Least squares

Let $X \in \mathbb{R}^{n \times p}$ be a design matrix.[1] The estimate minimises

$$L(\beta) = \sum_{i=1}^{n} \left(y_i - x_i^\top \beta\right)^2, \tag{1}$$

and its variance is $\mathrm{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}$. The density is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

The mean is $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ and $\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e$.

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{otherwise.} \end{cases} \qquad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Roots such as $\sqrt[3]{x+1}$, and $\alpha \leq \beta \neq \gamma$ with $x_{i_j} \neq y'$.

[1] We assume $n > p$.